

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

BASIC MATH AND CALCULUS

LECTURE-4

PHYSICS

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Topics To Be Covered

- 1 Revision
- 2 Trigonometry
- 3 Equation
- 4 Important Question
- 5 ..

Lecture no ③

Recap From 1st Lec.

Maths

- Maths Bohat Aasaan hai
- number
- $+$ $-$ $\times$ $\div$
- Fraction
- Angle / Power
- Trigonometry
- Graph / Equation
- $\frac{dy}{dx}$ $\int y dx$

Fraction

- $\frac{1}{2} = 0.5$
- $\frac{1}{3} = 0.33$
- $\frac{1}{4} = 0.25$
- $\frac{1}{10} = 0.1$
- $\frac{1}{100} = 0.01$
- $5.42 \times 100 = 542$

Power

- $2^2 = 2 \times 2$
- $x^0 = 1$
- $x^a \cdot x^b = x^{a+b}$
- $(x^a)^b = x^{a \times b}$

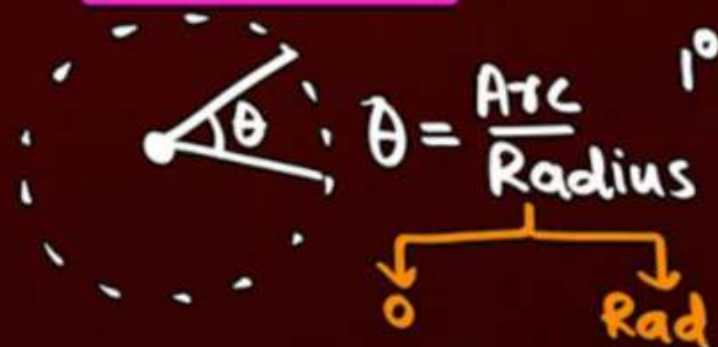
$$\frac{1}{\infty} = 0$$

$$\sqrt{2} = 2^{\frac{1}{2}}$$

$$\sqrt{2} = 1.4$$

$$\sqrt[3]{3} = 1.7$$

Angle θ



$$\rightsquigarrow 180^\circ = \pi \text{ rad}$$

$$1^\circ = \frac{\pi}{180} \text{ rad}$$

$$1 \text{ rad} = \frac{180^\circ}{\pi}$$

Recap From 2nd Lec.

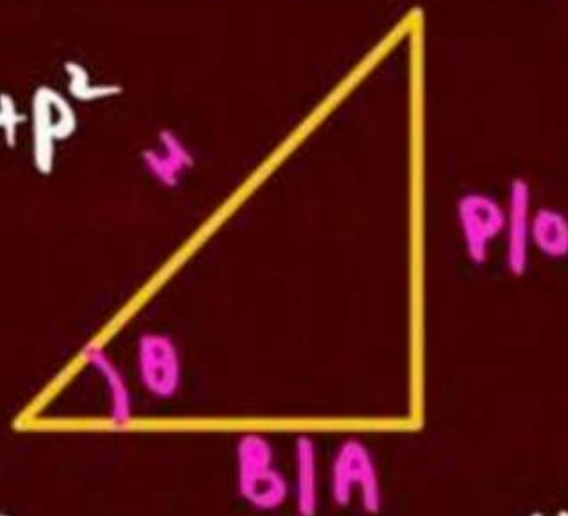
Trigonometry Table

	0°	30°	45°	60°	90°	120°	135°	150°	180°
Angle	0°	30°	45°	60°	90°	120°	135°	150°	180°
$\sin \theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0	$-\frac{1}{2}$	$-\frac{1}{\sqrt{2}}$	$-\frac{\sqrt{3}}{2}$	-1
$\tan \theta$	$\frac{0}{1}=0$	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	$\frac{1}{0}=\infty$	$-\sqrt{3}$	-1	$-\frac{1}{\sqrt{3}}$	0

Trigonometry

Special Δ

$H^2 = B^2 + P^2$



$\operatorname{cosec} \theta = \frac{1}{\sin \theta}$

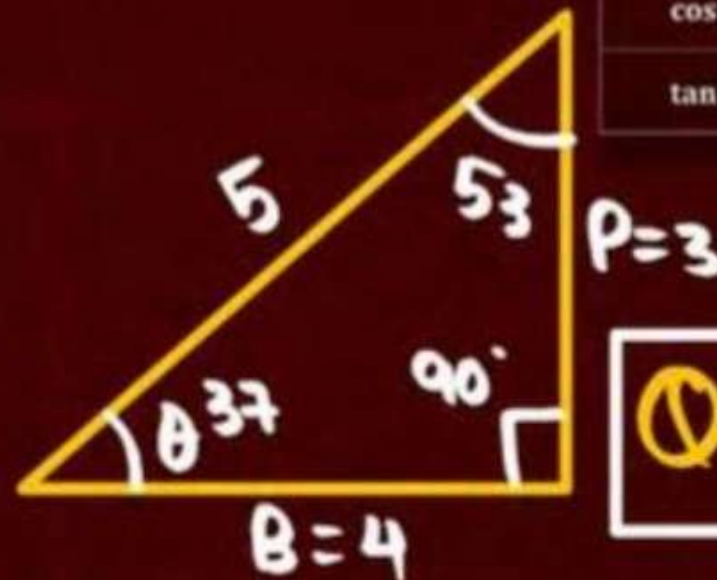
$\sec \theta = \frac{1}{\cos \theta}$

$\cot \theta = \frac{1}{\tan \theta}$

$\sin^2 \theta + \cos^2 \theta = 1$

$\sin \theta = \frac{P}{H}$
 $\cos \theta = \frac{B}{H}$
 $\tan \theta = \frac{P}{B}$

$\operatorname{cosec} \theta = \frac{H}{P}$
 $\sec \theta = \frac{H}{B}$
 $\cot \theta = \frac{B}{P}$



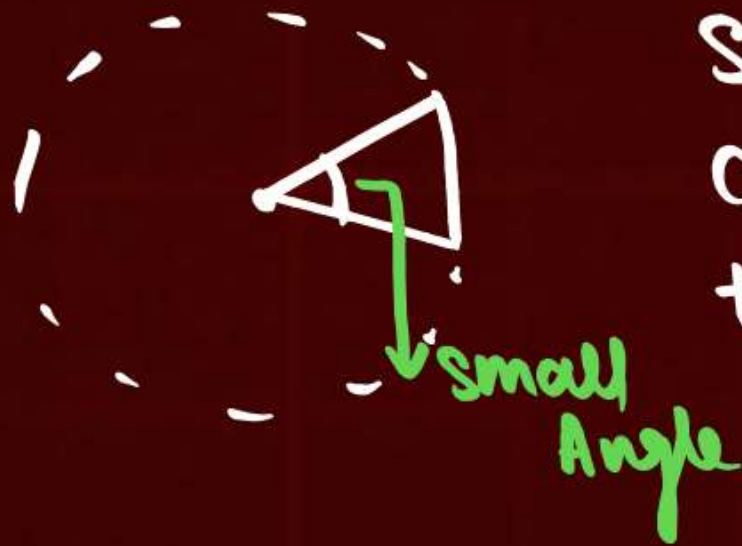
Quadrant System



Recap From 3rd Lec.



Small Angle Approx.



$$\sin \theta = \theta$$

$$\cos \theta = 1$$

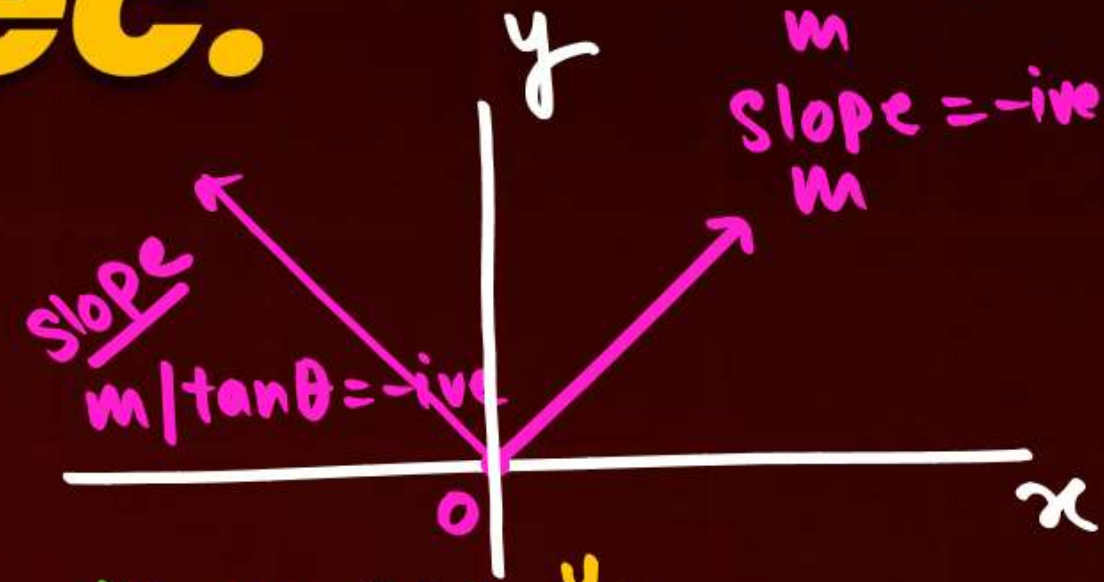
$$\tan \theta = \theta$$

Equation

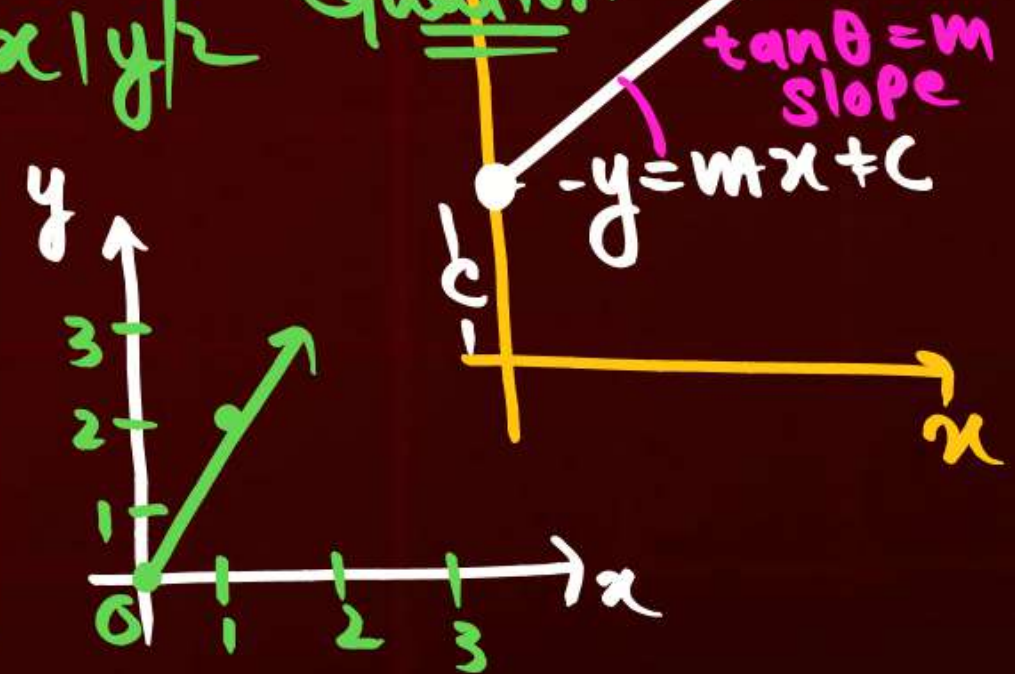
English Statement

Slope
 $y = \theta x$
 $\frac{y}{x} = \theta$

x	0	1	2
y	0	2	4



(i) Straight line equation

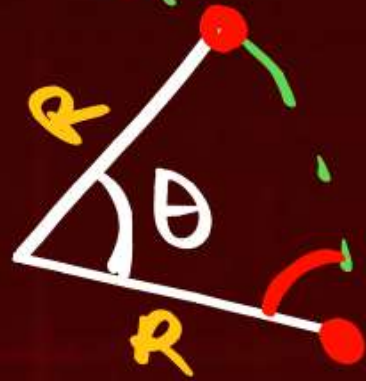


LOVE is the best feeling? NAH

Try a ROLEX on DAD's wrist and GUCCI
bag in MOM's hand...

Small Angle Approximation

- Angle :- Bohat Barra hota hai.



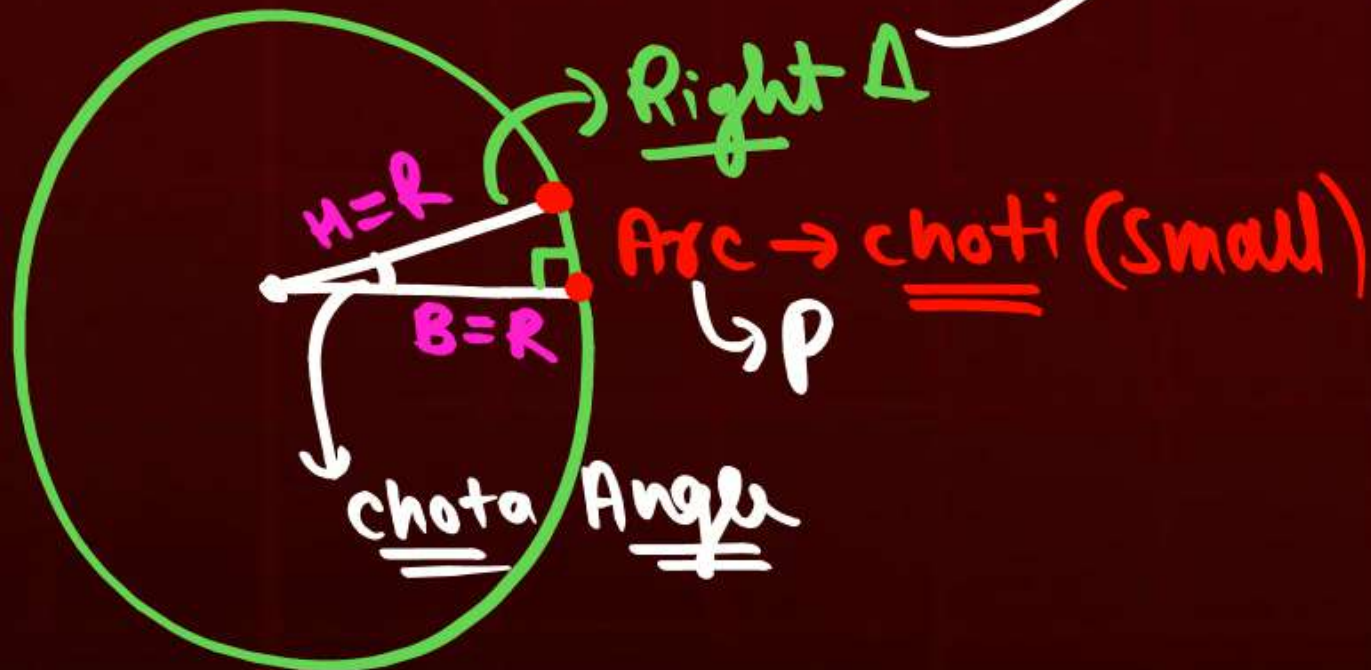
$$\theta = \frac{A}{R}$$

$\theta \rightarrow$ Small
1°, 2°

$$\underline{\sin \theta} = \frac{P}{H} = \frac{A}{R} = \theta \quad \checkmark$$

$$\underline{\cos \theta} = \frac{B}{H} = \frac{R}{R} = 1$$

$$\underline{\tan \theta} = \frac{P}{B} = \frac{A}{R} = \theta$$



QUESTIONS

Cos 2° is equal

- 1 1
- 2 2
- 3 3
- 4 5

$$\frac{\cos(2^\circ)}{\cos 2^\circ} \rightarrow \text{small angle}$$

$$= 1$$

$$\sin(2^\circ)$$

$$= \sin 2^\circ$$

$$= 2^\circ$$

$$= 2 \times 1^\circ$$

$$= \cancel{2} \times \frac{\pi}{\cancel{90} \cancel{180}} \text{ rad} = \frac{\pi}{90} \text{ rad}$$

$$\begin{aligned}
 \underline{\underline{Q}} \quad & \underline{\underline{\tan 2^\circ}} \\
 & \downarrow \\
 & 2^\circ \\
 & 2 \times 1^\circ \\
 & \cancel{12} \times \frac{\pi}{90 \cancel{180}} \text{ rad}
 \end{aligned}$$

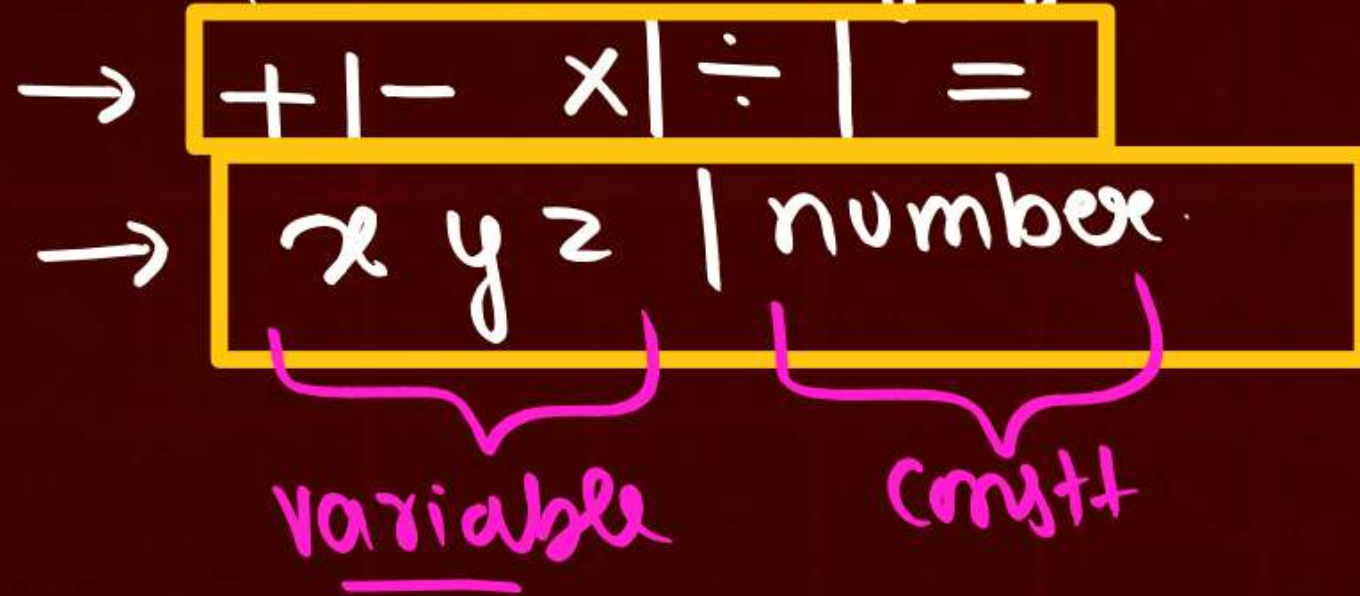
$$\frac{\pi}{90} \text{ rad}$$

$$\begin{aligned}
 \underline{\underline{\tan 1^\circ}} \\
 & \downarrow \\
 & 1^\circ \\
 & \downarrow \\
 & \frac{\pi}{180} \text{ rad}
 \end{aligned}$$

$$\begin{aligned}
 \underline{\underline{Q}} \quad & \sin 1^\circ \\
 & \downarrow \\
 & 1^\circ \\
 & \downarrow \\
 & \frac{\pi}{180} \text{ rad} \\
 & \underline{\underline{=}}
 \end{aligned}$$

Equation

- when you write an English statement in Maths language



Q The cost of Pen^y is twice the cost of pencil^x?

Sol Let the cost of Pen = y
cost of pencil = x

ATQ $\text{COP} = 2 \text{COP}$

$y = 2x$

x	0	1	2	3
y	$y = 2x$ $y = 2 \times 0$ $= 0$	$y = 2x$ $= 2 \times 1$ $= 2$	$y = 2x$ $= 2 \times 2$ $= 4$	$y = 2x$ $= 2 \times 3$ $= 6$

Graph

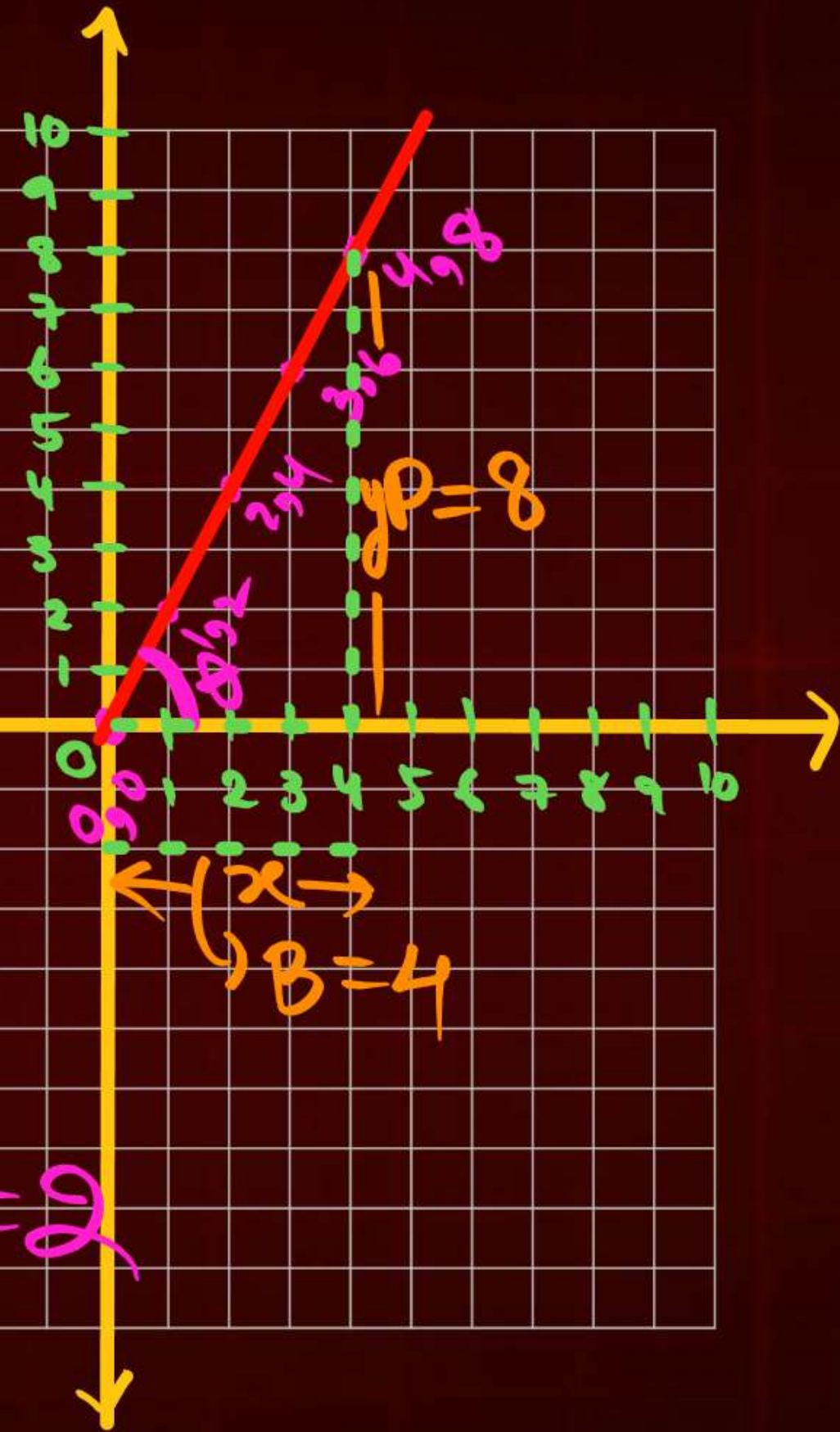
$$y = 2x$$

$$\text{Slope} = \tan \theta = m = 2$$

x	0	1	2	3	4
y	0	2	4	6	8

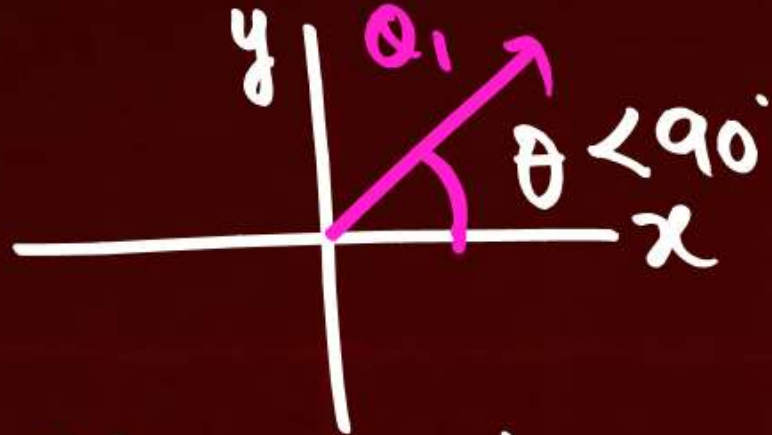
Slope → Kitni charrayi
Kitni utrai hai

$$\text{Slope}(m) = \frac{y}{x} = \frac{P}{B} = \tan \theta = \frac{P}{B} = \frac{8}{4} = 2$$

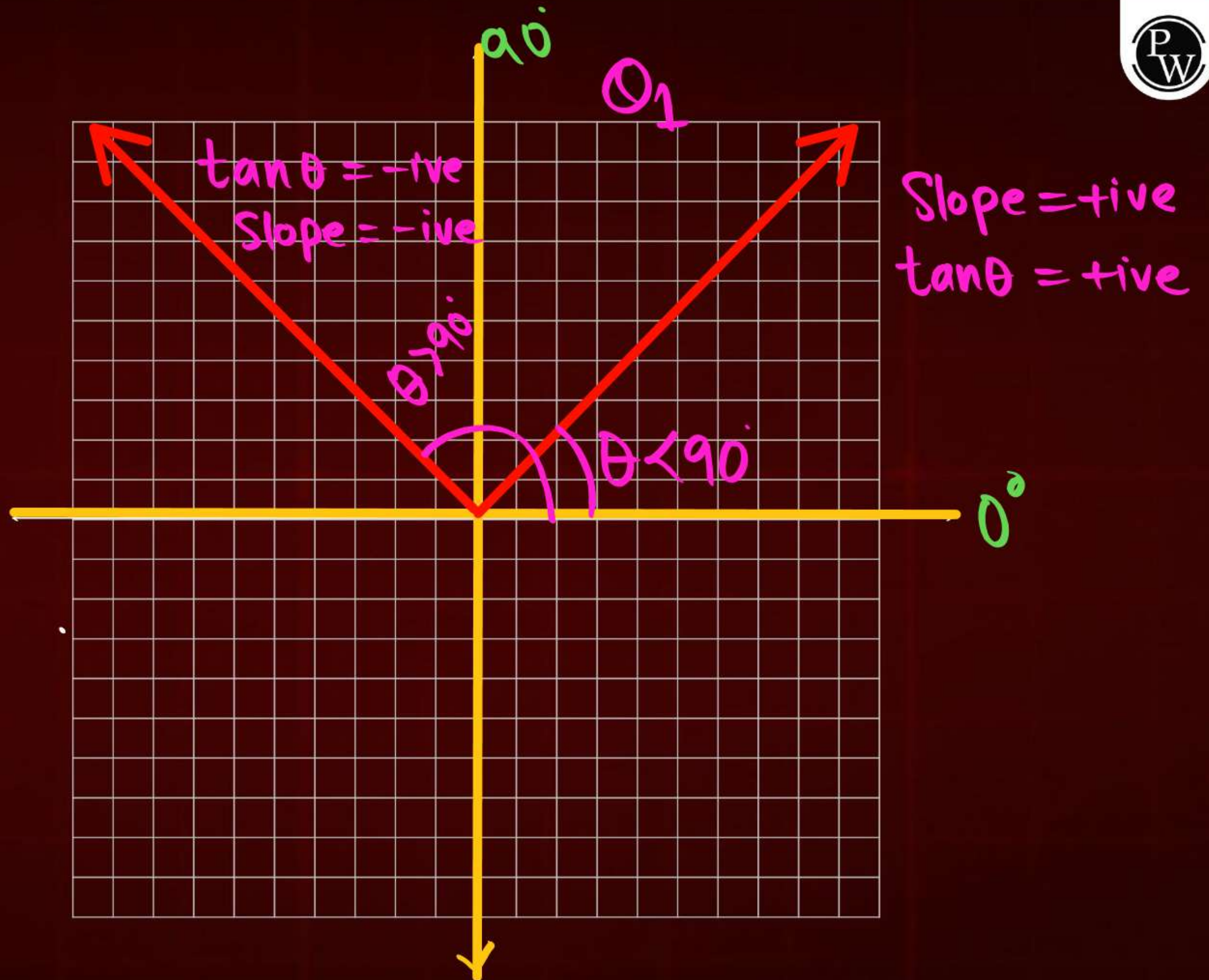
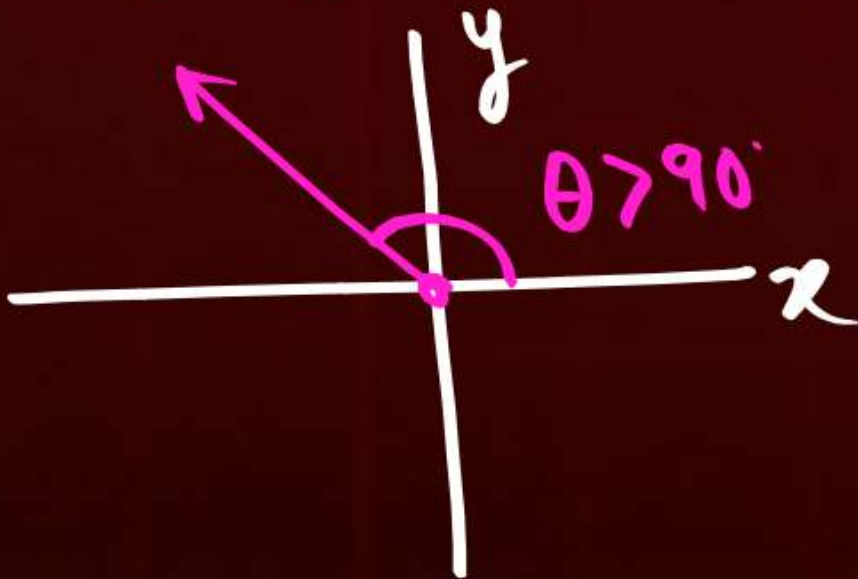


Graph

- Slope = +tive
Tan θ = +tive



- Slope = -ive
tan θ = -ive



- $y = \underline{2}x + \underline{2}$ → Intercept (y-intercept)

Slope

 $y = 2 \times 0 + 2$
 $y = 0 + 2 = 2$

↓
y → value

 $x = 0$

x	0	1	2
y	$y = 2x + 2$ $= 2 \times 0 + 2$ $= 0 + 2$ $= 2$	$y = 2x + 2$ $= 2 \times 1 + 2$ $= 2 + 2$ $= 4$	$y = 2x + 2$ $= 2 \times 2 + 2$ $= 4 + 2$ $= 6$

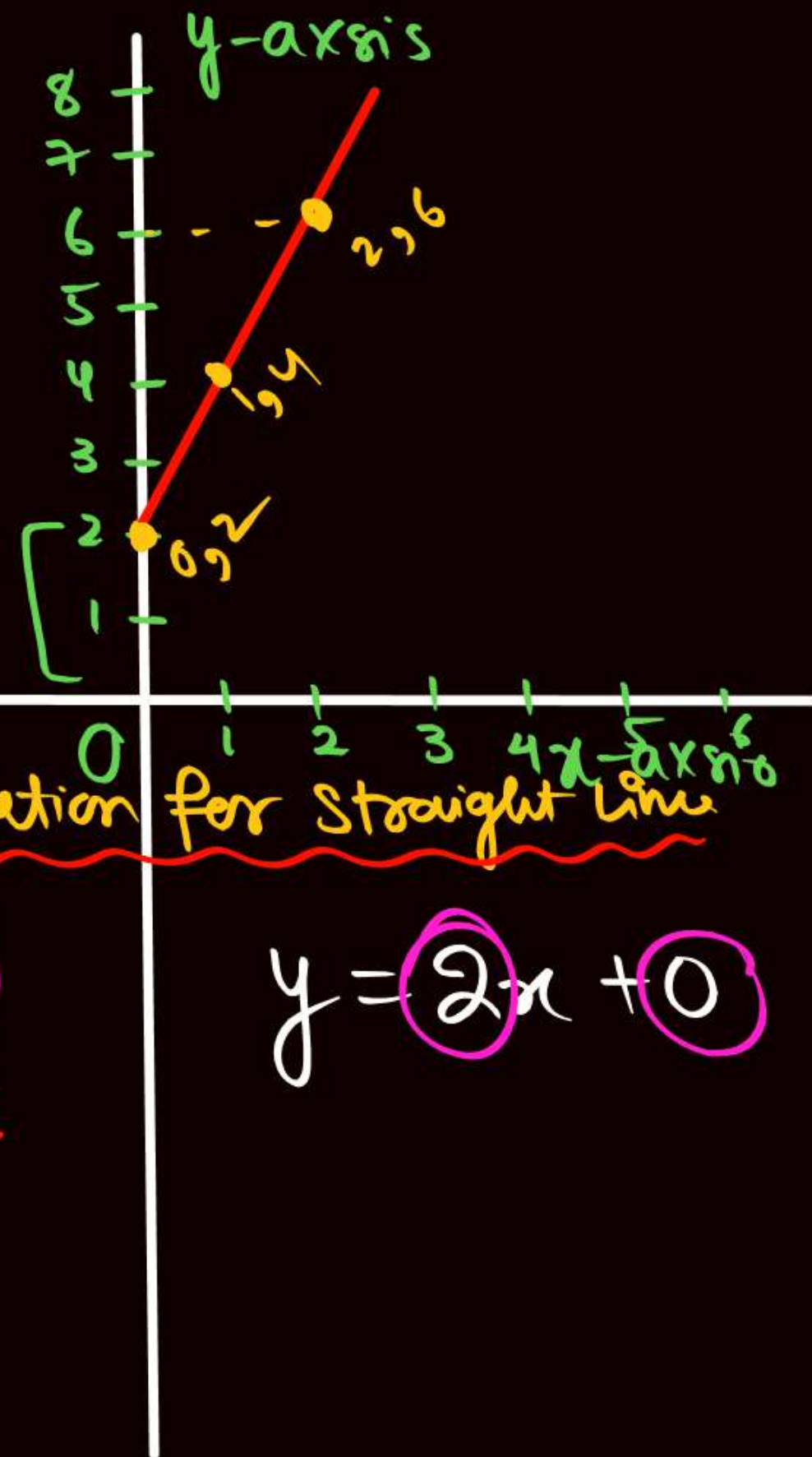
$$2 = \text{Intercept (c)}$$

General equation for Straight line

$$y = 2x + 2$$

$$\rightarrow y = mx + c$$

$$y = 2x + 0$$



Equation $\longrightarrow$ Table $\longrightarrow$ Graph

$$y = mx + c$$

$$y = x^2$$

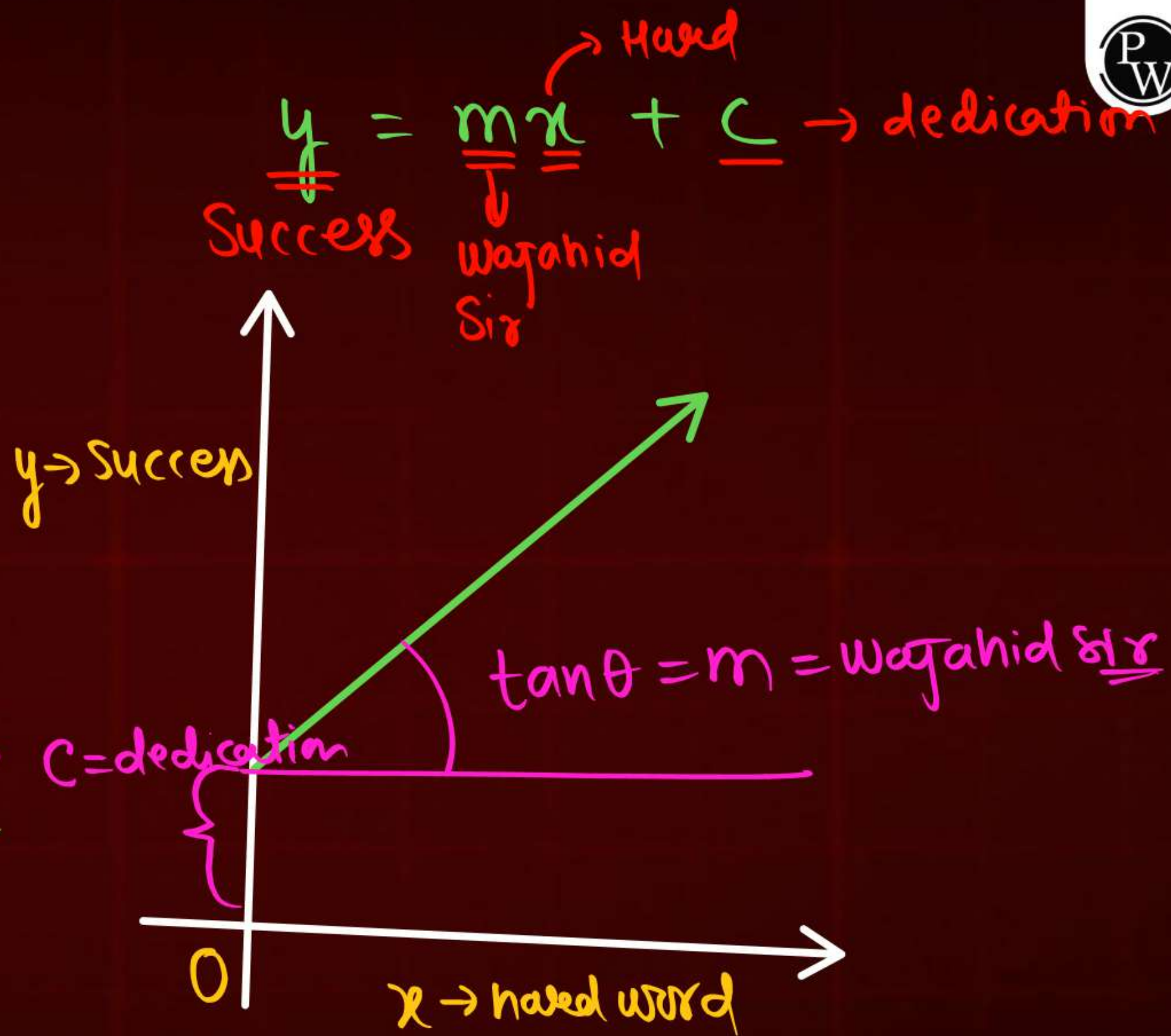
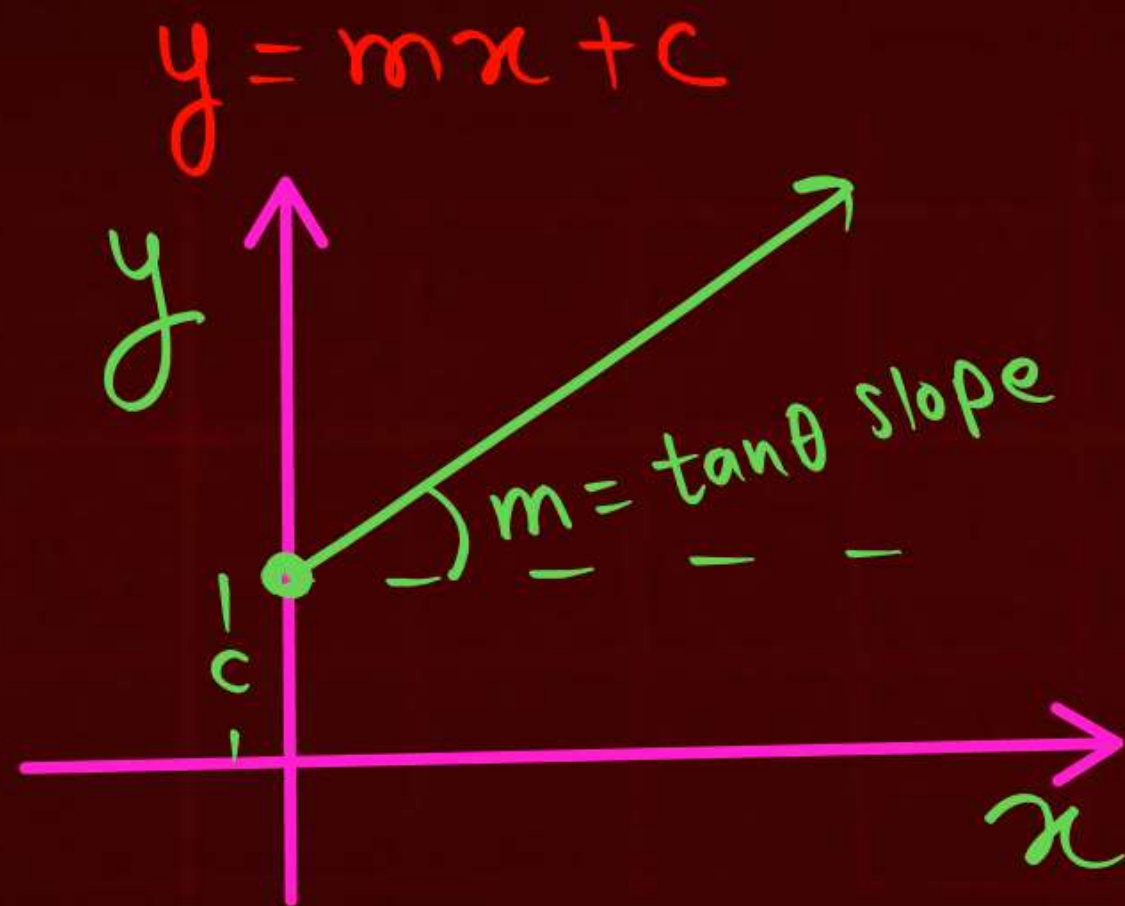
$$y = -x^2$$

$$x^2 + y^2 = R^2$$

* Graph $\longrightarrow$ depend on equation

Graph

1. Straight line graph

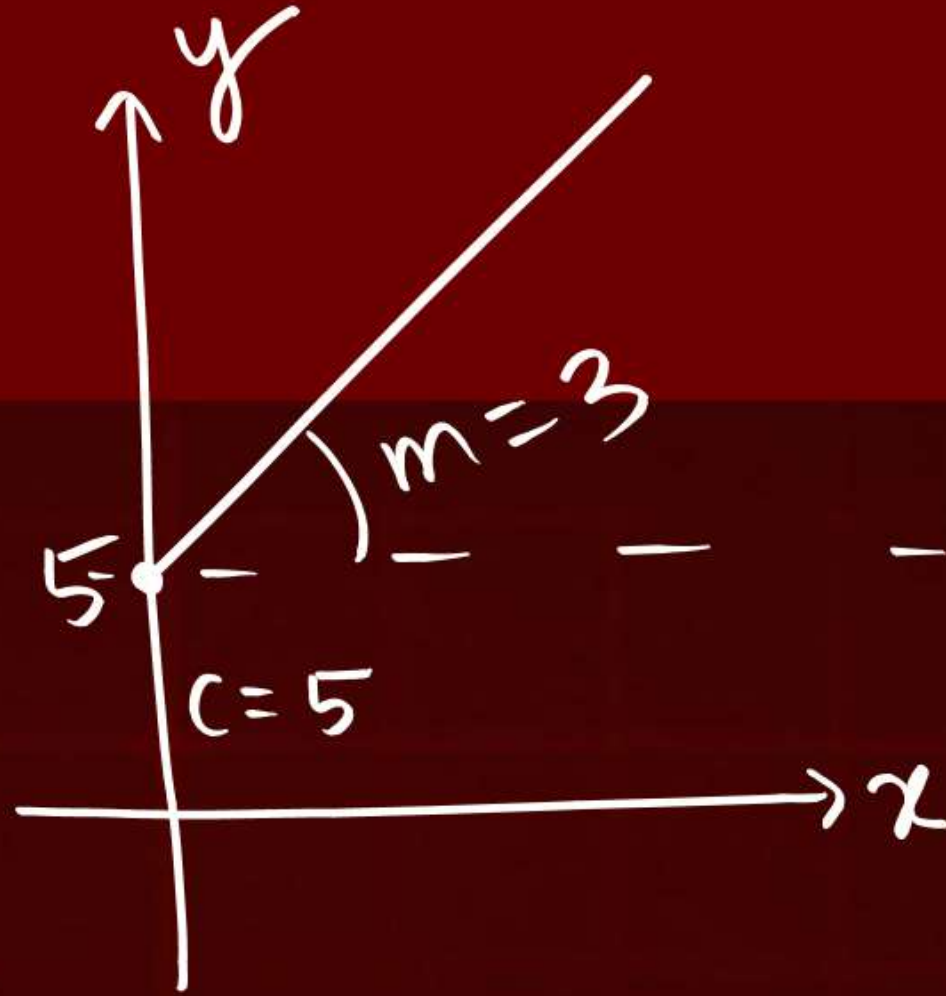


QUESTIONS

$$y = \underline{3}x + \underline{5} \quad \checkmark$$

$$m = 3$$

$$c = 5$$

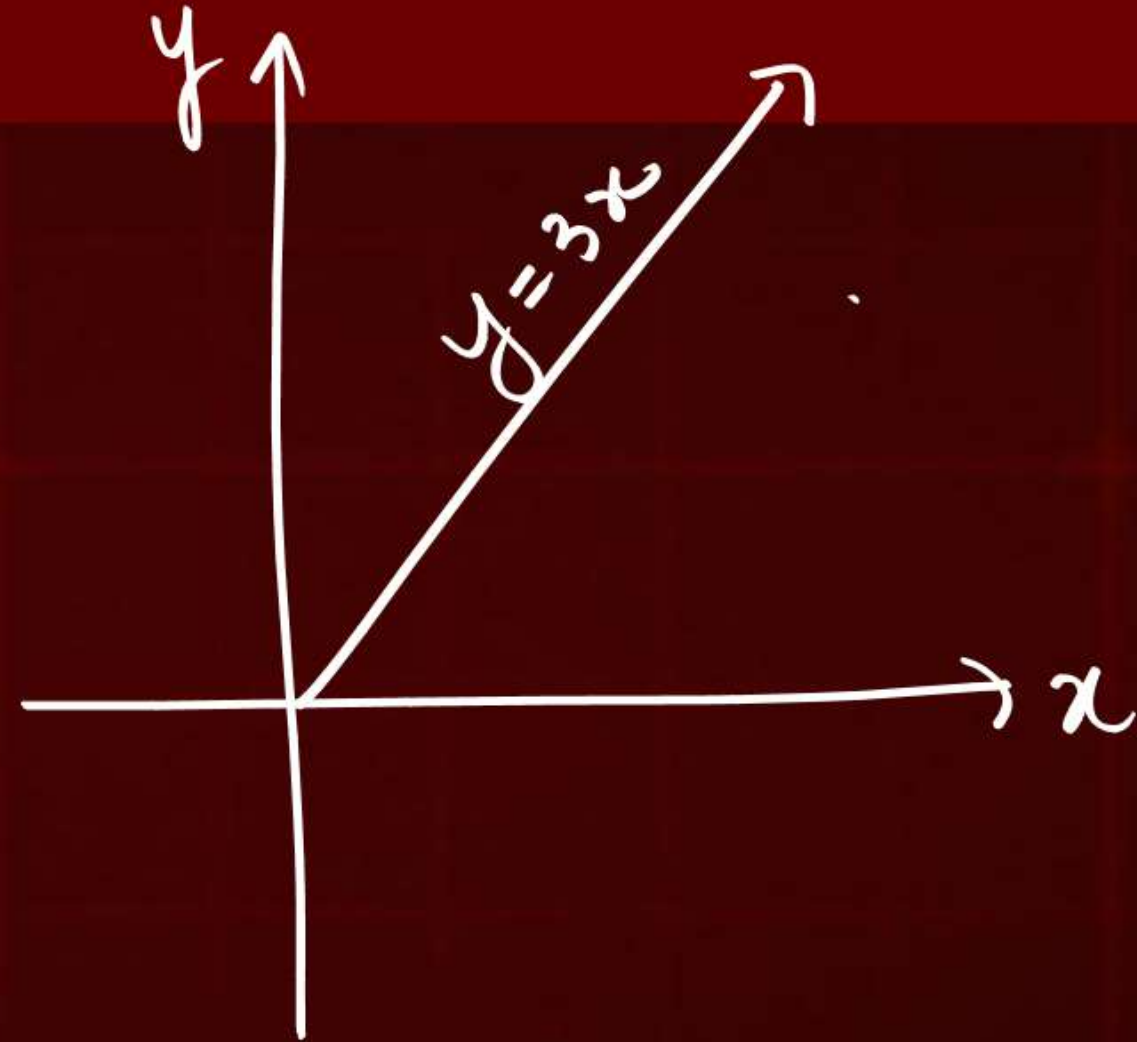


QUESTIONS

$$y = 3x, \quad C = 0$$

$$y = 3x + 0$$

$$m = 3 \quad c = 0$$



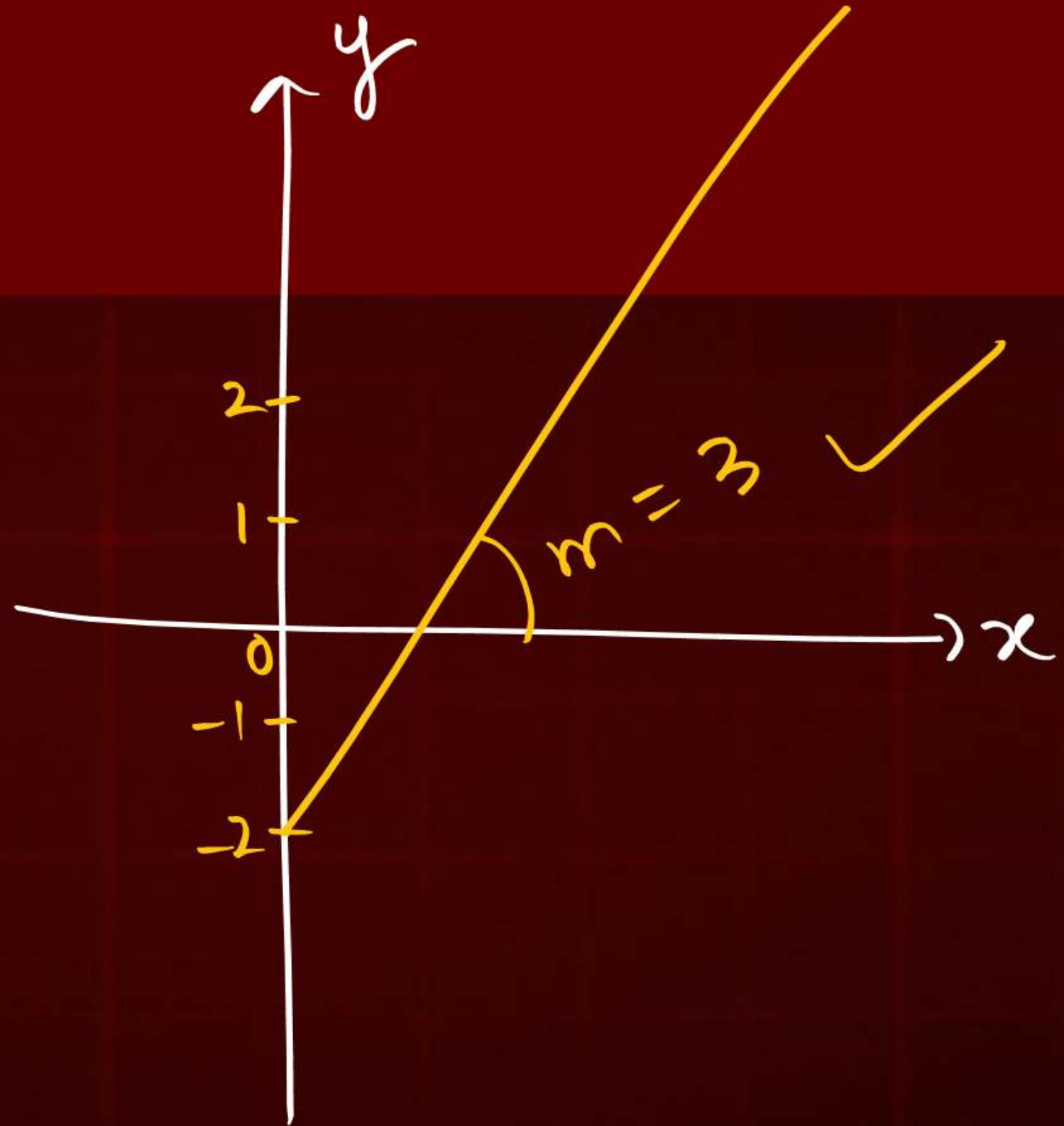
QUESTIONS

$$y = 3x - 2,$$

$$y = 3x - 2$$

$$y = \underline{3}x + -2$$

$$m=3 \quad c=-2$$

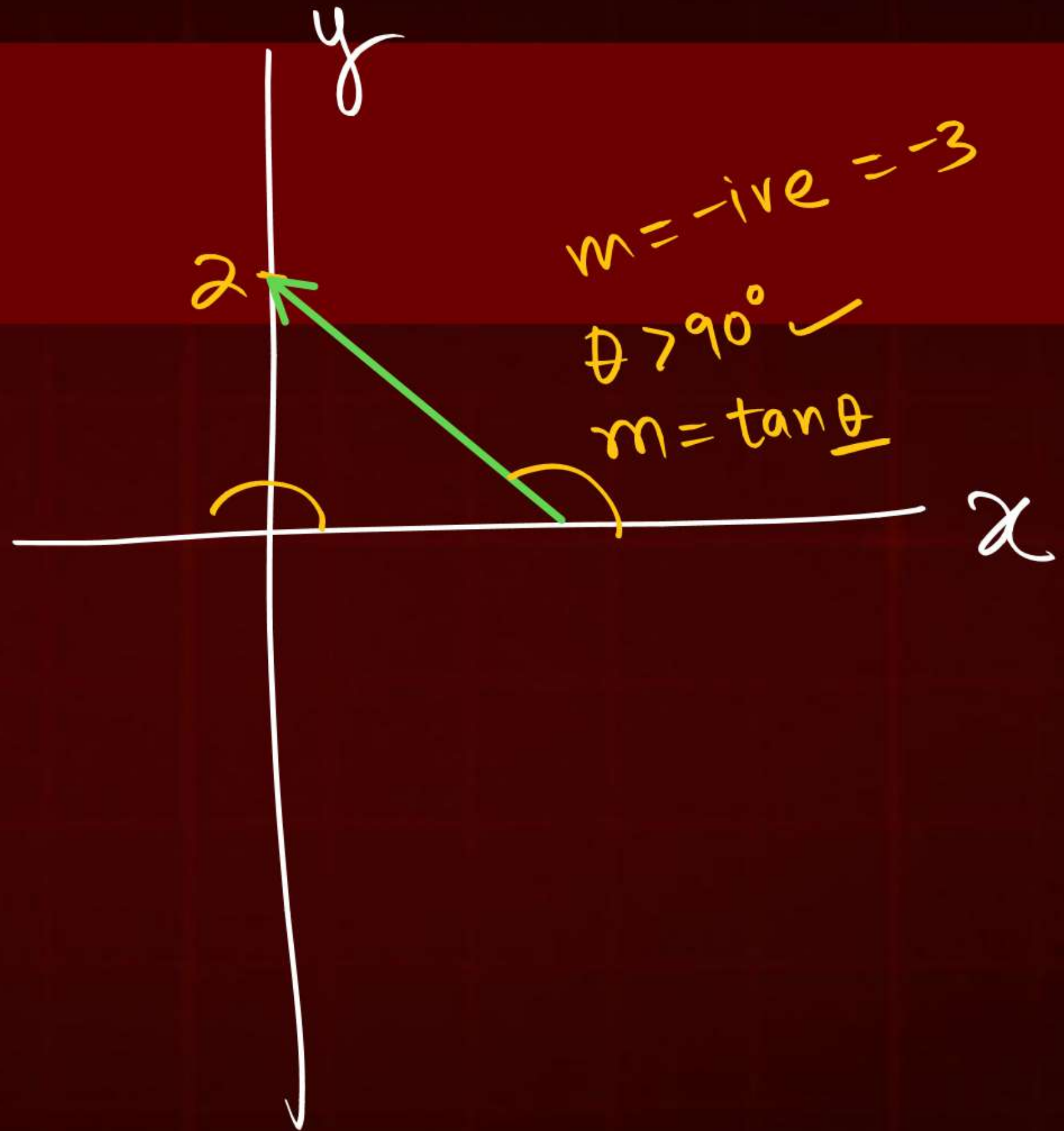


QUESTIONS

$$y = -3x + 2,$$

$$m = -3$$

$$c = 2$$



QUESTIONS

$$y = \underline{-3}x - 3,$$

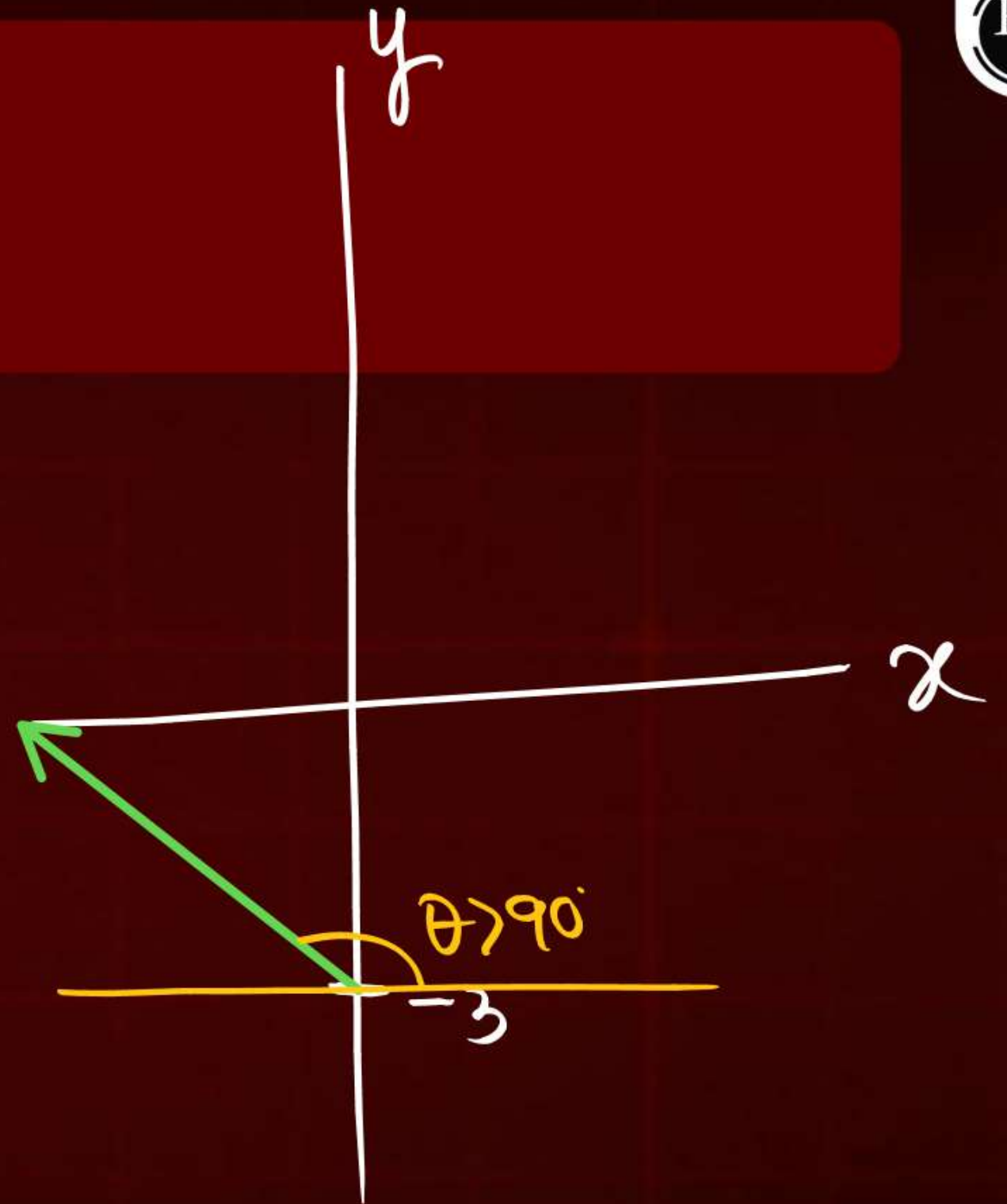
$$m = -3$$

$$c = -3$$

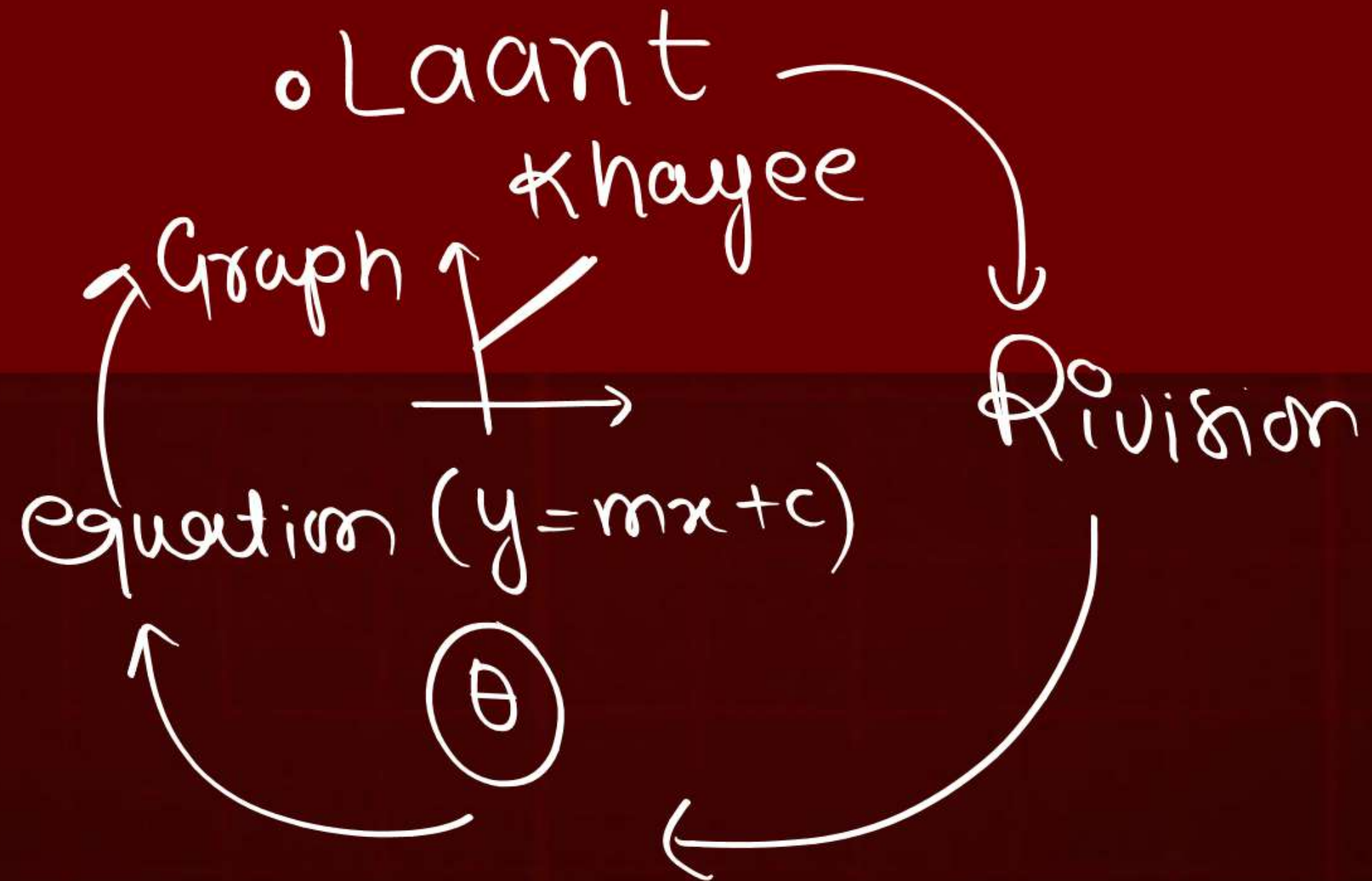
$$\begin{aligned} y &= -3x - 3 \\ &= \underbrace{-3x} + \underbrace{-3} \end{aligned}$$

$$= m = -3$$

$$c = -3$$



SUMMARY



HOMEWORK



Notes → Community

$$\check{y} = 2(x) + 5$$



Thank
You